

DIRECT TO: ABEAM points, CRS IN and CRS OUT

The three DIRECT TO options of the A380 FCOM, on the shared flight plan code

Branch	feature/a380/mfd/direct-to-options
Aircraft	A380X (flight plan functions shared with the A32NX FMGC)
Commits	f98743703 feat(a380x/mfd): DIRECT TO with ABEAM points, CRS IN and CRS OUT
Base	pr/10-a380x-mfd-fcom-pages (first built on the old fork master 280800cf7)
Status	Merged in develop. 10 unit tests; harness-checked; not yet flown.

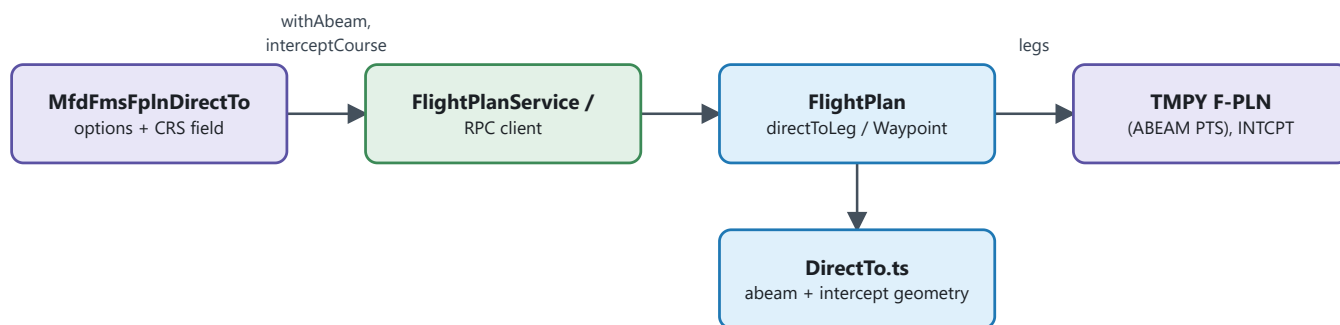
Purpose

The DIRECT TO page showed DIRECT WITH ABEAM, CRS IN and CRS OUT, but they could not be selected (no back end).

Specification (A380 FCOM DSC-22-FMS-20-30, DIRECT TO)

- **DIRECT WITH ABEAM:** for each bypassed waypoint, an abeam point on the direct leg, named AB + 5 characters of the waypoint; the temporary F-PLN shows "(ABEAM PTS)".
- **CRS IN / CRS OUT:** a course entered into / out of the target (format NNN or NNNT, FCOM defaults). The FMS computes an intercept point (INTCPT) on that course and flies direct to it, then along the course. No intercept point when the course cannot be intercepted (more than 160 deg, parallel or behind): the leg is a course to / from the target.
- The flight plan functions are shared (`fmgc/.../plans/DirectTo.ts`): `abeamPointOnLeg`, `directToInterceptPoint`, new leg factories and a pending-abeam flag cleared when the TMPY is inserted.

 Component added or changed by the feature MSFS / simulator Aircraft system (systems host, FMS, FWS) Display / page
 Data, LVar, storage Crew, flypad, external program



Architecture

A CI (course to intercept) leg was tried first: FBW's Geo.legIntercept picks a bogus start point (NaN). A computed INTCPT waypoint is used instead. Not done: the HDG-engaged / NAV-armed variant and the ADJUST DESIRED TRK OR HDG message; the intercept point is not recomputed when the track changes.

PR description (from the PR draft)

From `pr-drafts/14-direct-to-options.md`, section "Summary of Changes" onwards.

Summary of Changes

The DIRECT TO page showed the three options of the A380 FCOM (DSC-22-FMS-20-30, DIRECT TO) but they could not be selected.

- **DIRECT WITH ABEAM:** an abeam point on the direct leg for each bypassed waypoint, named AB + 5 characters of the waypoint; the temporary F-PLN shows "(ABEAM PTS)".
- **CRS IN / CRS OUT:** a course into / out of the target (entry format NNN or NNNT, FCOM default courses). The FMS computes an intercept point (INTCPT) on that course, flies direct to it and then along the course. When the course cannot be intercepted (more than 160 deg, parallel or behind), the leg is a course to / from the target without an intercept point.
- The flight plan functions are shared (`fmgc/.../plans/DirectTo.ts`: abeam point on a leg, intercept point), with 10 unit tests.

Cockpit API Changes

None.

Screenshots (if necessary)

TO ADD: the DIRECT TO page with each option and the resulting TMPY F-PLN and ND.

References

- A380 FCOM DSC-22-FMS-20-30, DIRECT TO page and its options.

Additional context

A CI leg was tried first: FBW's `Geo.legIntercept` returns a bogus start point (NaN) for it, so a computed intercept waypoint is used. Not done: the HDG-engaged / NAV-armed variant, the ADJUST DESIRED TRK OR HDG message, recomputing the intercept when the track changes.

Discord username (if different from GitHub): **TO ADD**

Testing instructions

1. DIRECT TO a waypoint further along the plan WITH ABEAM: abeam points AB... in the TMPY, "(ABEAM PTS)".
2. CRS IN 090 to a waypoint ahead: INTCPT then the waypoint; CRS OUT: the waypoint then the course.
3. A course behind or parallel: no INTCPT, course to / from the target.